



USRW @ RECSYS 2026

When Search and Recommendation become One Conversation

This presentation is based on the collective work and publications of numerous contributors at Spotify.

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Outline



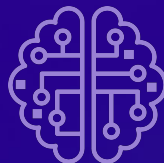
Personalization

From curation to reasoning.



One Conversation

When search and recommendation converge.



Building It

Generative models, agents, and what we learned.



Looking Ahead

Open challenges and opportunities.

Personalization

From curation to reasoning.

Personalization: the shared challenge

Search, recommendations, feeds — today's core surfaces are personalized.



Abundance

Catalogs contain far more content than anyone can navigate. Both search and recommendation exist to solve this, differently.



Scale

Results are assembled for each person, in milliseconds, across billions of users. Search and recommendation have mostly tackled this separately.



Stake

What gets surfaced shapes how people discover culture, information, and commerce. Those decisions involve users, creators, and platforms, each with different interests.

How personalization has evolved

Human Curated

- Humans decide what is relevant and surface it.
- **Example:** *Indie Chill*, hand-picked playlists by music editors.



Predictive

- Models learn from behaviour to predict and rank content.
- **Example:** *Discover Weekly*, a weekly mixtape unique to each listener.



Generative

- Systems understand intent and reason over it.
- **Example:** "Find jazz podcasts for my commute", on-demand, intent-driven results.





One Conversation

When search and recommendation converge.

Search and recommendations: two ways to connect people to content

User intent	SEARCH Search: expresses explicit, query-driven intent	RECOMMENDATIONS Recommendations: express implicit, preference-driven intent
System approach	SEARCH Search: matches a query to a corpus and ranks by relevance	RECOMMENDATIONS Recommendations: model taste over time and surface what might resonate
Interaction model	SEARCH Search: is user-initiated and single-turn	RECOMMENDATIONS Recommendations: are system-initiated and continuous



A long time ago...

Information retrieval and filtering: two sides of the same coin.

— Belkin & Croft, 1992



Thirty years later, **AI is dissolving this boundary.**

Search and recommendation are converging.

The convergence is now empirical

A survey of information retrieval and recommender systems in the era of large language models: Two sides of the same task

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Same roles

Across **450+ papers**, search and recommendation independently converge on three shared LLM roles: **engine**, **re-ranker**, and **generator**.

A paradigm shift

LLM-based work outnumbers traditional approaches **14:1** in search and **7:1** in recommendation.

Joint search & recommendation architectures surged exponentially from 2023.

Different velocities

Search converges faster via native textual data.
Recommendation transitions from behavioral signals to language.

Both converge toward the same **generative endpoint**.

“We conclude that the distinction between retrieval and recommendation is becoming a historical artifact and call for the formal recognition of a new, integrated science of information discovery.”

Search and Recommendation as One Conversation

Search

"find something specific"

Recommendations

"surprise me"

Multi-turn

"refine, follow up, continue"

The Agent

decide what to do

Intent

What is the user trying to accomplish?

Profile

What do we know about this user?

Context

What happened earlier in this session?

The Generative Core

generate the response

World knowledge

Platform data



Building It

Generative models, agents, and what we learned.

Building the next generation of personalization

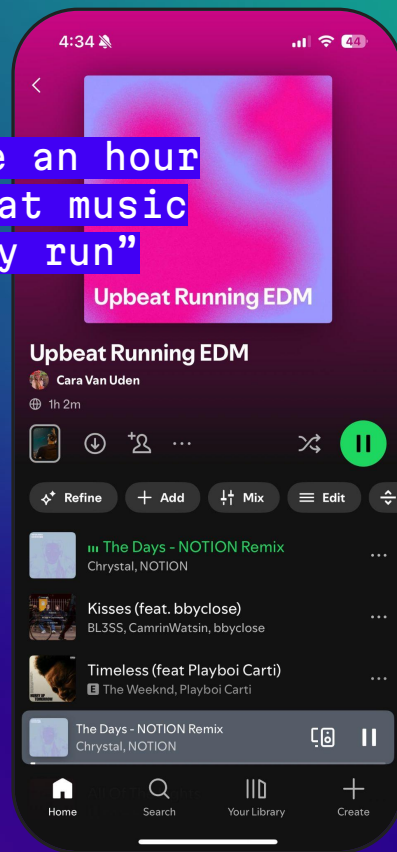
A unified model — One model for recommendation and search.

Steerable recommendations — Familiar or discovery, controlled at request time.

Agentic search — The system decides whether to search or recommend.

Conversational agents — Multi-turn dialogue, bootstrapped from zero.

“Give me an hour
of upbeat music
for my run”



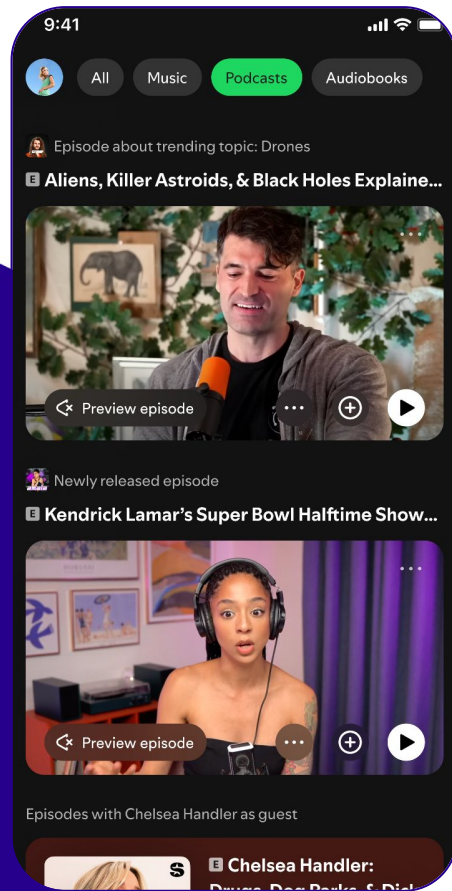
From Ranking to Generating

One conversation requires one model rather than separate systems for search and recommendations.

A Unified Model for Personalization:

Language-Steerable Generative Recommendation, Search, and User Understanding — KDD 2026

From Habits to Discovery: Deploying LLMs for Personalized Generative Recommendations at Spotify — KDD 2026



A Unified Model (I)

THE PROBLEM

Separate models cannot share representations across heterogeneous content, nor handle the full range of what one conversation requires.

THREE CAPABILITIES IN ONE MODEL

Recommendation

Generate personalized recommendations across content types.

Text Retrieval

Retrieve catalog items from natural-language queries.

Reasoning & Explanations

Describe user preferences and explain why something was recommended.

KEY DESIGN DECISIONS

Interface

Natural language as the interface. All capabilities are accessed through natural language.

Representation

Semantic IDs, not text titles. Items are generated as token sequences, keeping them grounded in actual catalog.

A Unified Model (II)

THREE-STAGE TRAINING

Stage 1 — Semantic Foundation

LLM learns to map catalog items to token sequences, without losing its language competence.



Stage 2 — Domain Grounding

Aligns the model with catalog structure and user behavior signals.



Stage 3 — Capability Induction

Multi-task training across recommendation, retrieval, and user understanding.

OFFLINE EVALUATION

+57%

HR@10 episodes

+46%

HR@10 audiobooks

+47%

HR@10 text retrieval*

Steerable Recommendations (I)

THE PROBLEM

One conversation may call for familiar content or new discovery. Recommenders trained on listening history cannot serve both without retraining.

THREE-STAGE TRAINING

Stage 1 — Semantic Foundation

Learns Semantic IDs token sequences for all podcast episodes.



Stage 2 — Instruction Tuning

Model learns to follow the control token [familiar vs unfamiliar].

ARCHITECTURE

Domain-adapted language model

A LLM fine-tuned on Spotify's podcast catalog to generate episodes as Semantic ID sequences.

Control token [familiar / unfamiliar]

Appended at serving time to steer what gets recommended.

Steerable Recommendations (II)

PRODUCTION DEPLOYMENT

Offline training

Language model trained on Spotify's podcast catalog with Semantic IDs.



Candidate generation

Model outputs token sequences, decoded back to episode IDs.



Control at request time

Control token appended per request, with no separate model branch or retraining needed.

ONLINE A/B RESULTS

+5.4%

non-habitual streams

+14.3%

new-show streams

What we learned

01

Shared representations

Recommendation, retrieval, and explanation all operate in the same semantic ID space. Built apart, they cannot.

02

One model, different experiences

The same model can serve familiar listening or new discovery, chosen at request time. No separate model for each.

03

Domain adaptation vs. language competence

Teaching the model a new domain risks losing the reasoning abilities it needs. Staged training helps, but we are not there yet.

04

Beyond familiar vs. discovery

Familiar vs. discovery is just one dimension. There is much more, content hypotheses, taste profiles, and beyond.

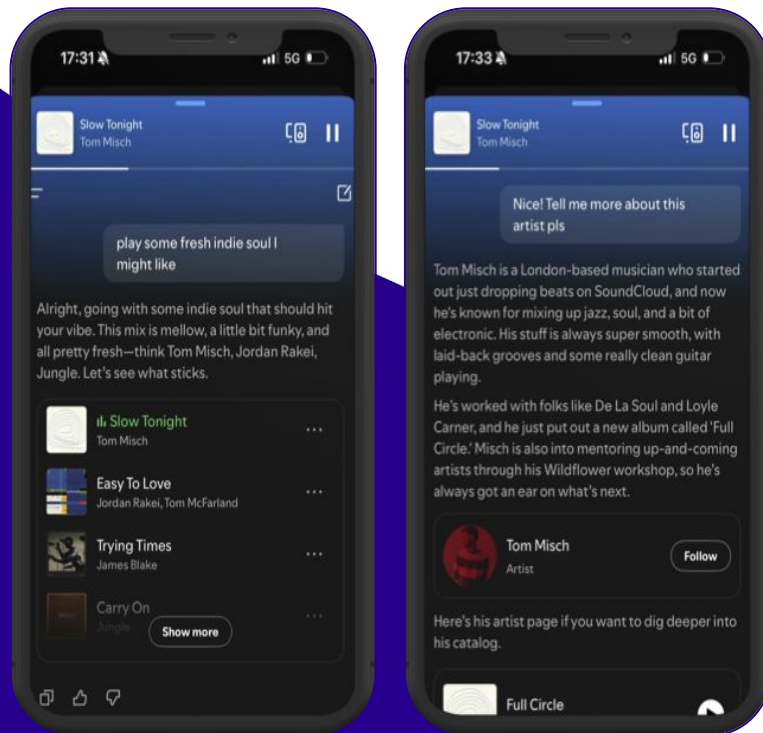
From Search to Agent

One conversation means deciding whether to search, recommend, or engage in dialogue, and being capable of all three.

You Say Search, I Say Recs:

A Scalable Agentic Approach to Query Understanding and Exploratory Search at Spotify — RecSys 2025

Bootstrapping Conversational Recommendation Agents at Spotify: Synthetic Data Generation and Self-Improvement Loops — RecSys 2026

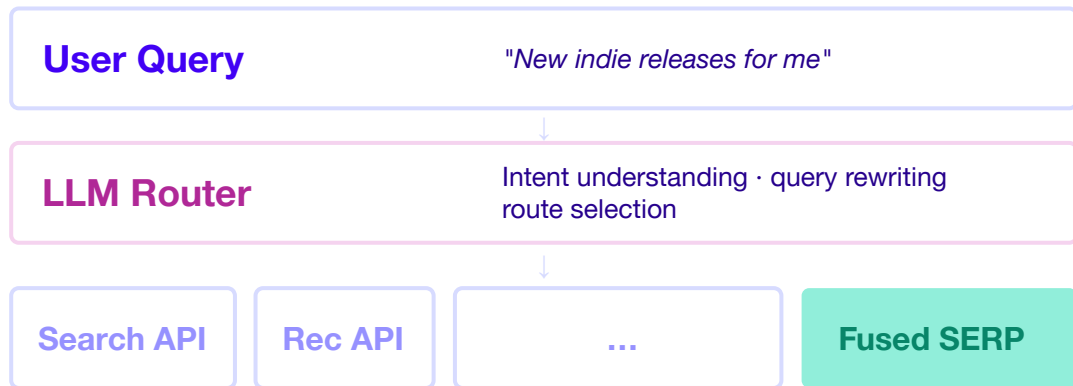


Agentic Search (I)

THE PROBLEM

Users express what they want in natural language. The agent's job is to understand what they mean and act on it.

PARALLEL FUSION ROUTER (PFR)



KEY DESIGN DECISIONS

Routing

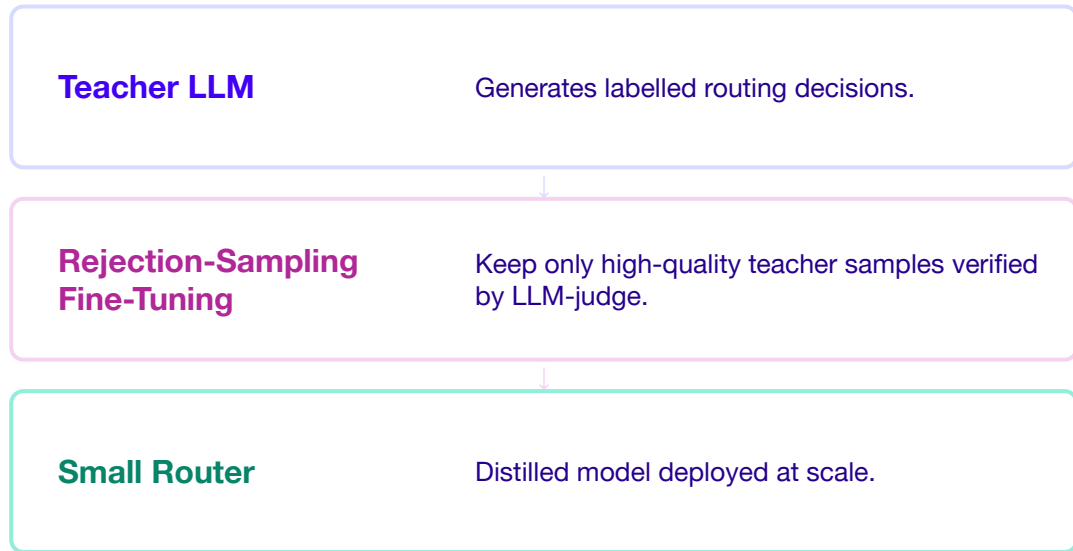
Route only on query. Keeps latency low and caching efficient at scale.

Execution

Parallel tool calls. Search API, Rec API, and sub-agents run in parallel; total latency is that of the slowest call, not the sum.

Agentic Search (II)

MAKING THE ROUTER PRODUCTION-READY



SMALL ROUTER VS. TEACHER LLM

-60% latency

-99% cost

+3% quality

ONLINE A/B RESULTS

+115%
similar artist queries

+91%
new music releases

+25%
broad music searches

+15%
broad podcast searches

Conversational Agents (I)

THE PROBLEM

A multi-turn agent needs conversation data to train on. At cold start, that data does not exist.

SYNTHETIC DATA PIPELINE

Single-turn Seed

Real queries/interactions

Dialog Plan

LLM generates a script per capability

Simulated Conversations

Simulated user + live agent

Quality Filtering

LLM judge scores each conversation per capability

CAPABILITIES

Content Refinement

"Actually, more upbeat"

Instruction Retention

"Keep the jazz, add vocals"

Anaphora Resolution

"More like the third one"

Intent Shift

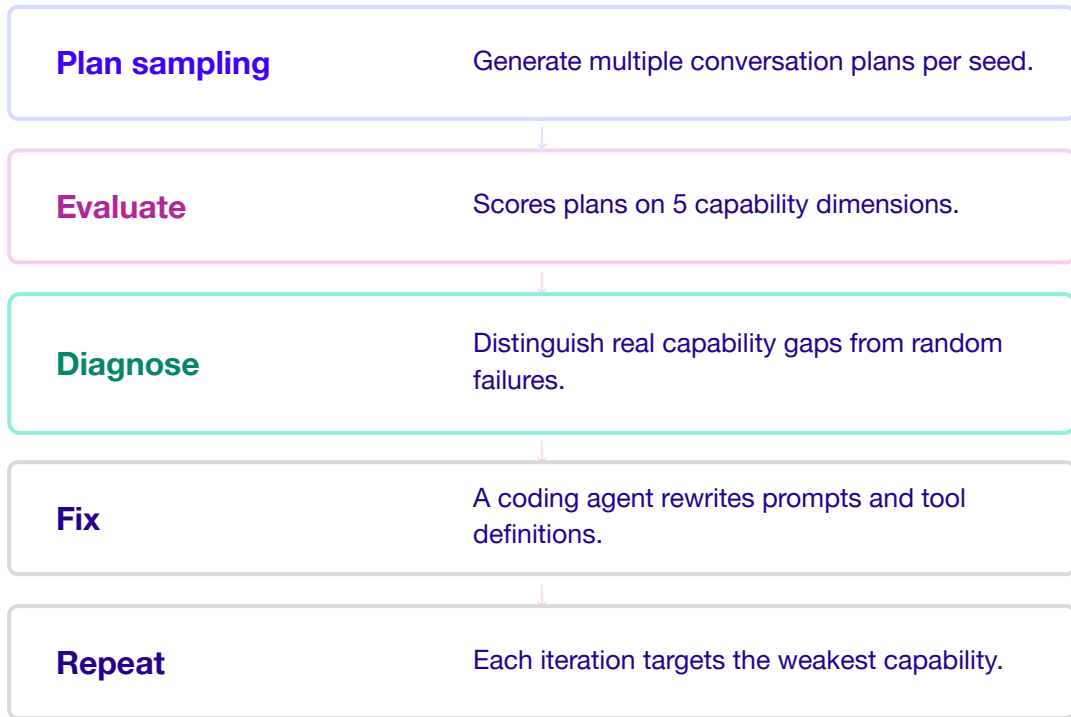
"Now find me a podcast"

Separate Input

"Play that and also queue X"

Conversational Agents (II)

SELF-IMPROVEMENT LOOP



ONLINE A/B RESULTS

+14%
user listening

-5%
skip rate

+5%
weekly active users

+8%
improve pass@1 (offline).

What we learned

01

Intent first

Everything downstream (routing, retrieval, generation) depends on getting the user's intent right.

02

Synthetic data for cold start

Synthetic generation can bootstrap a new domain, capability, or conversation type without waiting for real data.

03

From capable to reliable

Agents can hold conversations. Making them reliably good across all intents, including unseen ones, is an open problem.

04

Evaluation by dimension

Defining the right capability dimensions is what makes LLM-as-judge work. Multiple judges and automated prompt optimization let it scale.

Looking Ahead

Open challenges and opportunities.

Let us recap

What we built

- A unified model for recommendation, retrieval, and explanation.
- Steerable recommendations without retraining.
- Agentic search with intent-aware routing.
- Conversational agents bootstrapped from zero.

What we are advancing

- Search and recommendation are converging into one conversation.
- Shared representations with semantic IDs bind the pieces.
- The retrieval vs. recommendation distinction is becoming a historical artifact.

Our work through the convergence lens

Three LLM roles — Biancofiore, Palma, Pomo et al., Computer Science Review, 2027

ROLE 01

Embedded Engine

The LLM is the backbone, directly outputting item IDs or semantic tokens.

Unified model with Semantic IDs for recommendation, retrieval, and explanation.

ROLE 02

Semantic Re-ranker

The LLM works on top of an earlier retrieval stage, using context to judge relevance.

Agentic search router deciding whether to search, recommend, or both.

ROLE 03

Response Generator

The task adapts to the model, from constrained generation to open-ended conversation.

Steerable recommendations via control tokens; conversational agents bootstrapped from synthetic data.

Open Problems

These are today's open problems — as frontier models evolve, the answers and the questions will shift.

CHALLENGE 01

Generalization without retraining

The foundations for new content, new intents, and new tasks are here.

Extending reliably to what the model has not seen remains open.

CHALLENGE 02

Grounding & factual fidelity

A generative system can surface items that do not exist.

RAG helps, but guaranteeing factual fidelity in unified systems is open.

CHALLENGE 03

Evaluation for unified systems

LLM-as-judge works for individual capabilities.

Evaluating a full conversation across search, recommendation, and dialogue requires new frameworks.



Thank You

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This presentation is based on the collective work and publications of numerous contributors at Spotify.